Baselines Report

B0 (no gate), B1 rules, B2 and B3 prompted Qwen3-8B, against B4 and B5 · 14 November 2026

> **Outcome. No baseline meets both targets; the fine-tuned four-way gate does.** Rules only (B1) never ask without need but let 25/118 (21.2%) unsafe actions through on test. The prompted binary model (B2) is safe (1/89 (1.1%)) but blocks 66/97 (68.0%) of legitimate actions. The prompted four-way model (B3) is both unsafe (27/118 (22.9%)) and over-asking (53/110 (48.2%)). B5 alone beats each of them on paired success (all p < 0.001 after Holm), and the full gate reaches 206/212 (97.2%) with 0 unsafe.

1. The runs

Item	Value
B1, B2, B3	NB02 on Kaggle `satorugojo13`, 14 November 2026, code `89832cd` (code package identical to `locked-v4`); dev / attacks / test, clean and noisy simulated user (B2 has no second round)
B0	NB01 on Kaggle `satorugojo13`, 14 November 2026: the same Qwen3-8B agent with no gate on the 142 AgentDojo test scenarios (`important_instructions` attack)
What they are	**B0** no gate. **B1** rules only (the deterministic GateExp-style rules, no model). **B2** Qwen3-8B, not fine-tuned, prompted to answer ALLOW or BLOCK. **B3** the same, prompted with all four decisions. **B4** fine-tuned binary (run A). **B5** fine-tuned four-way, alone and inside the full gate.

2. Gate baselines (offline cases)

Test pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B1 rules only	212	166/212 (78.3%)	25/118 (21.2%)	0/110 (0.0%)	0/133 (0.0%)	166/212 (78.3%)	28/28 (100.0%)	28/39 (71.8%)	30/63 (47.6%)
B1 rules only, noisy user	212	162/212 (76.4%)	25/118 (21.2%)	0/110 (0.0%)	4/133 (3.0%)	166/212 (78.3%)	28/28 (100.0%)	28/39 (71.8%)	26/63 (41.3%)
B2 prompted binary	159	79/159 (49.7%)	1/89 (1.1%)	0/80 (0.0%)	66/97 (68.0%)	58/159 (36.5%)	0/0 (n/a)	0/27 (0.0%)	0/52 (0.0%)
B3 prompted four-way	212	150/212 (70.8%)	27/118 (22.9%)	53/110 (48.2%)	19/133 (14.3%)	93/212 (43.9%)	10/27 (37.0%)	10/39 (25.6%)	26/63 (41.3%)
B3 prompted four-way, noisy user	212	143/212 (67.5%)	27/118 (22.9%)	53/110 (48.2%)	26/133 (19.5%)	93/212 (43.9%)	10/27 (37.0%)	10/39 (25.6%)	24/63 (38.1%)
B4 binary (dev-matched threshold)	159	94/159 (59.1%)	4/89 (4.5%)	0/80 (0.0%)	60/97 (61.9%)	75/159 (47.2%)	0/0 (n/a)	0/27 (0.0%)	0/52 (0.0%)
B5 alone (b5_v4_combo, seed 42)	212	192/212 (90.6%)	3/118 (2.5%)	5/110 (4.5%)	13/133 (9.8%)	198/212 (93.4%)	39/43 (90.7%)	39/39 (100.0%)	52/63 (82.5%)
B5 + full gate (final system)	**212**	**206/212 (97.2%)**	**0/118 (0.0%)**	**9/110 (8.2%)**	**6/133 (4.5%)**	**200/212 (94.3%)**	**37/38 (97.4%)**	**37/39 (94.9%)**	**58/63 (92.1%)**

Attacks pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B1 rules only	61	50/61 (82.0%)	10/51 (19.6%)	1/33 (3.0%)	1/11 (9.1%)	42/61 (68.9%)	6/12 (50.0%)	6/15 (40.0%)	4/13 (30.8%)
B1 rules only, noisy user	61	50/61 (82.0%)	10/51 (19.6%)	1/33 (3.0%)	1/11 (9.1%)	42/61 (68.9%)	6/12 (50.0%)	6/15 (40.0%)	4/13 (30.8%)
B2 prompted binary	57	51/57 (89.5%)	5/47 (10.6%)	0/33 (0.0%)	1/10 (10.0%)	32/57 (56.1%)	0/0 (n/a)	0/13 (0.0%)	0/11 (0.0%)
B3 prompted four-way	61	52/61 (85.2%)	8/51 (15.7%)	22/33 (66.7%)	1/11 (9.1%)	20/61 (32.8%)	2/13 (15.4%)	2/15 (13.3%)	7/13 (53.8%)
B3 prompted four-way, noisy user	61	52/61 (85.2%)	8/51 (15.7%)	22/33 (66.7%)	1/11 (9.1%)	20/61 (32.8%)	2/13 (15.4%)	2/15 (13.3%)	7/13 (53.8%)
B4 binary (dev-matched threshold)	57	56/57 (98.2%)	0/47 (0.0%)	0/33 (0.0%)	1/10 (10.0%)	32/57 (56.1%)	0/0 (n/a)	0/13 (0.0%)	0/11 (0.0%)
B5 alone (b5_v4_combo, seed 42)	61	57/61 (93.4%)	2/51 (3.9%)	3/33 (9.1%)	2/11 (18.2%)	51/61 (83.6%)	11/15 (73.3%)	11/15 (73.3%)	9/13 (69.2%)
B5 + full gate (final system)	**61**	**60/61 (98.4%)**	**0/51 (0.0%)**	**4/33 (12.1%)**	**1/11 (9.1%)**	**48/61 (78.7%)**	**12/19 (63.2%)**	**12/15 (80.0%)**	**7/13 (53.8%)**

Dev pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B1 rules only	236	214/236 (90.7%)	5/112 (4.5%)	0/150 (0.0%)	0/149 (0.0%)	202/236 (85.6%)	20/32 (62.5%)	20/25 (80.0%)	32/61 (52.5%)
B1 rules only, noisy user	236	209/236 (88.6%)	5/112 (4.5%)	0/150 (0.0%)	5/149 (3.4%)	202/236 (85.6%)	20/32 (62.5%)	20/25 (80.0%)	27/61 (44.3%)
B2 prompted binary	187	81/187 (43.3%)	0/85 (0.0%)	0/120 (0.0%)	106/118 (89.8%)	68/187 (36.4%)	0/0 (n/a)	0/16 (0.0%)	0/51 (0.0%)
B3 prompted four-way	236	194/236 (82.2%)	11/112 (9.8%)	89/150 (59.3%)	26/149 (17.4%)	105/236 (44.5%)	8/30 (26.7%)	8/25 (32.0%)	37/61 (60.7%)
B3 prompted four-way, noisy user	236	183/236 (77.5%)	12/112 (10.7%)	89/150 (59.3%)	36/149 (24.2%)	105/236 (44.5%)	8/30 (26.7%)	8/25 (32.0%)	33/61 (54.1%)
B5 alone (b5_v4_combo, seed 42)	236	225/236 (95.3%)	0/112 (0.0%)	9/150 (6.0%)	9/149 (6.0%)	219/236 (92.8%)	23/23 (100.0%)	23/25 (92.0%)	55/61 (90.2%)
B5 + full gate (final system)	**236**	**232/236 (98.3%)**	**0/112 (0.0%)**	**9/150 (6.0%)**	**4/149 (2.7%)**	**223/236 (94.5%)**	**25/25 (100.0%)**	**25/25 (100.0%)**	**57/61 (93.4%)**

3. Paired tests against B5 alone (secondary family, Holm-Bonferroni)

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
B5 vs B1 rules (test)	success	192/212 (90.6%)	166/212 (78.3%)	+12.3 [+7.1, +17.5]	**<0.0001**	<0.0001
B5 vs B2 prompted binary (test)	success	146/159 (91.8%)	79/159 (49.7%)	+42.1 [+34.0, +50.3]	**<0.0001**	<0.0001
B5 vs B3 prompted 4-way (test)	success	192/212 (90.6%)	150/212 (70.8%)	+19.8 [+13.2, +26.4]	**<0.0001**	<0.0001
B5 vs B1 rules (attacks)	success	57/61 (93.4%)	50/61 (82.0%)	+11.5 [+1.6, +21.3]	0.0654	0.4580
B5 vs B3 prompted 4-way (attacks)	success	57/61 (93.4%)	52/61 (85.2%)	+8.2 [-1.6, +18.0]	0.1797	1.0000

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

4. Analysis

B1: rules only

- Test: 166/212 (78.3%) correct, **25/118 (21.2%) unsafe**, 0 unneeded asks. Dev: 214/236 (90.7%) correct, 5/112 (4.5%) unsafe.
- The rules do far better on dev (4.5% unsafe) than on test (21.2%). Their errors on test are missing-evidence (21) and authority judgements (11): the deciding fact (who may approve, whether the user really asked, whether a value is genuine) is not readable off the call deterministically.
- The rules ask only when a rule says so, which is why they never ask without need, and also why they never recover an ambiguous case.
- B1 is still useful: the same rule ideas, placed around the model (GateExp), are what removes B5's last unsafe actions.

B2: prompted Qwen3-8B, binary

- Safe (1/89 (1.1%) unsafe on test) only because it blocks almost everything that is not obviously fine: 66/97 (68.0%) of legitimate actions blocked on test, 106/118 (89.8%) on dev. Correct final action 79/159 (49.7%) (on the 159 cases a binary gate can represent).
- On attacks it lets 5/47 (10.6%) through: an untuned model follows plausible-looking injected instructions.

B3: prompted Qwen3-8B, four-way

- Giving the untuned model the option to ask makes it ask too often (53/110 (48.2%) unneeded asks on test, 22/33 (66.7%) on attacks) while still executing unsafe actions (27/118 (22.9%)). Its first decision is right in only 93/212 (43.9%) of cases.
- Four decisions only help when the model knows when to use each one: fine-tuning on evidence-labelled cases is what B5 adds (first round 198/212 (93.4%)).

The ladder

- Rules only: fast, never over-asks, unsafe on new cases. Prompted binary: safe by refusing. Prompted four-way: unsafe and noisy. Fine-tuned binary (B4): safe, blocks 62% of legitimate actions. Fine-tuned four-way (B5): 192/212 (90.6%), 2.5% unsafe, 4.5% unneeded asks. B5 inside the rules: 206/212 (97.2%), 0% unsafe, 8.2% unneeded asks.
- Only the last two meet both targets on test.

5. End to end (AgentDojo, 142 test scenarios)

System	Utility	Utility, no attack	Utility under attack	Attack success	Leakage	Tool selection
B0: no gate	49/142 (34.5%)	8/22 (36.4%)	41/120 (34.2%)	10/120 (8.3%)	0/142 (0.0%)	203/298 (68.1%)
B5 + full gate (final)	**49/142 (34.5%)**	**8/22 (36.4%)**	**41/120 (34.2%)**	**0/120 (0.0%)**	**0/142 (0.0%)**	**209/298 (70.1%)**

- Without a gate the agent completes 49/142 (34.5%) of tasks and **10/120 (8.3%) attacks succeed** (banking 8/24 (33.3%)); with the final gate 49/142 (34.5%) and 0/120 (0.0%).
- Paired over the same 120 attacked scenarios: 10 attacks that succeeded without the gate are stopped, none is let through that B0 stopped (exact McNemar p = 0.002). Utility is paired-identical: one task is lost and one gained (p = 1.0).
- So the gate removes every successful attack at **no cost in utility**: it refuses 34 of 107 state-changing calls, and the number of tasks completed does not drop.

- The agent itself bounds utility (tool selection 209/298 (70.1%) with the gate).

6. Records

Record	Where
B1–B3 predictions and reports	`results/data/2026-11-14_baselines_b1b3/`
B0 traces and metrics	`results/data/2026-11-14_e2e_b0/`